

**INFORMATICS STUDY PROGRAM
CURRICULUM
UNIVERSITAS MULTIMEDIA
NUSANTARA**

Drafting Team

Rector of Universitas Multimedia Nusantara

Vice Rector I for Academic Affairs

Dean of the Faculty

Head of the Study Program

Deputy Head of the Study Program

Academic Information Bureau

CURRICULUM OF INFORMATICS STUDY PROGRAM

A. Profile

The Informatics Study Program at Universitas Multimedia Nusantara (UMN) offers a bachelor's degree with an option for a joint degree program in collaboration with Swinburne University of Technology. This program is designed for high school graduates with a science background or vocational school graduates with a background in information technology who aspire to build careers in artificial intelligence, cybersecurity, and digital interaction. UMN Informatics equips students with essential skills and knowledge for the dynamic IT industry, covering topics such as algorithms, programming, databases, artificial intelligence (AI), cybersecurity, and software development with a focus on digital interaction, including 3D games, augmented reality, and virtual reality.

The Informatics Study Program at UMN is currently undergoing international accreditation by ASIIN, which ensures the quality of education aligns with global standards. In addition, the program is also in the process of obtaining "Excellent" accreditation from LAM-INFOKOM, demonstrating UMN's commitment to maintaining high-quality education in the field of information technology. UMN places sustainability as a primary focus in both teaching and research, grounded in the principles of the Sustainable Development Goals (SDGs). A global approach is also strongly upheld, with collaboration with QS to expand international networks and enhance global reputation.

The Informatics Study Program continuously updates its laboratory facilities in the areas of artificial intelligence (AI), cybersecurity, and digital interaction, equipped with the latest hardware and software. Students also have the opportunity to obtain international professional

certifications that can expand their career prospects. Moreover, UMN offers various student mobility programs, such as the Merdeka Belajar (MBKM) program, student exchanges, and the opportunity to pursue a joint degree program with Swinburne University of Technology. All these facilities and programs are designed to provide a holistic learning experience, preparing Informatics students for global careers and contributing to technological advancement worldwide.

B. Career Prospects and Opportunities

The program opens doors to diverse career paths across industries, both locally and internationally.

1. Software & System Development

Graduates can pursue roles in software engineering and application development, including:

- Software Developer / Engineer
- Mobile and Web Application Developer
- Full-Stack Developer
- Game Developer (2D/3D, AR/VR)
- Embedded Systems Developer

2. Cybersecurity & IT Infrastructure

With specialized knowledge in security and systems, career options include:

- Cybersecurity Analyst
- Ethical Hacker / Penetration Tester
- System Administrator
- Network Security Specialist
- Cloud Infrastructure Engineer

3. Artificial Intelligence & Data Science

Students with a focus in AI and data analytics can work in roles such as:

- AI / Machine Learning Engineer
- Data Scientist
- Data Analyst
- Business Intelligence Developer
- Computer Vision Engineer

4. Digital Interaction & User Experience

Graduates may also contribute to digital product design and innovation through:

- UI/UX Designer
- Interaction Designer
- AR/VR Developer
- Digital Product Designer

5. Entrepreneurship & Innovation

With an entrepreneurial mindset, students can explore:

- Technopreneur / Startup Founder
- Product Manager (Technology Sector)
- Innovation Consultant
- Technology Business Strategist

6. Academic & Research Fields

For those interested in advancing knowledge and education:

- Informatics Researcher (AI, Cybersecurity, etc.)
- Academic Lecturer or Educator
- R&D Engineer (Research and Development)
- Policy Analyst in Tech and Innovation

C. Graduate Profiles

1. **Software Developer:** Become a proficient software developer capable of designing user-centric interfaces and scalable application architectures. Develop software solutions across various domains, including artificial intelligence and cybersecurity. Conduct thorough testing, monitoring, and performance evaluations to ensure functionality, security, and user experience. Analyze software behavior for continuous improvement and maintain long-term operability. Collaborate effectively within multidisciplinary teams to deliver complex, innovative technology projects.

2. **System Administrator:** Become a skilled system administrator responsible for managing and maintaining critical IT infrastructure within an organization. Oversee

the operation of computer systems, database management platforms, cybersecurity frameworks, and intranet networks. Ensure system reliability, data integrity, and secure access across platforms. Apply best practices in IT management to support operational efficiency and organizational continuity.

3. **Technopreneur:** Become a forward-thinking technopreneur capable of identifying and seizing business opportunities by leveraging technological advancements and expertise in Informatics. Develop innovative business models and technology-driven solutions to address market needs. Combine entrepreneurial insight with technical proficiency to create scalable, impactful ventures in the digital economy.

4. **Researcher:** Become a competent researcher who conducts and contributes to scientific inquiry in the field of Informatics. Engage in interdisciplinary research guided by ethical standards and scientific methodologies. Produce impactful findings that advance knowledge, inform industry practices, and support technological development in areas such as artificial intelligence, cybersecurity, and digital interaction.

D. Expected Learning Outcome

No.	ELO	Description
1	Ethical and Social Responsibility	Students are expected to understand and apply ethical principles and social considerations in technology-related decisions, ensuring responsible and sustainable professional conduct.
2	Problem Solving and Critical Thinking	Students should be able to analyze complex problems, evaluate alternatives critically, and develop

		effective, innovative solutions using logical reasoning.
3	Professional Skills (Entrepreneurial spirit in technology domain)	Students are expected to demonstrate initiative, creativity, and business acumen to identify opportunities and drive innovation within the technology sector.
4	Research and Innovation Mindset	Students should exhibit curiosity and an investigative approach, conducting research and generating innovative ideas to advance knowledge and technology.
5	Intelligent Systems Proficiency	Students are expected to design, develop, and apply intelligent systems using AI and machine learning techniques to address practical challenges.
6	Information Security Expertise	Students should be able to identify security risks, implement protective measures, and maintain the integrity, confidentiality, and availability of information systems.
7	Software Design and Implementation Abilities	Students are expected to design, develop, and maintain software solutions following best engineering practices and industry standards.

E. MBKM Program

1. Magang (Career Acceleration Program)

Magang is a structured program that allows students to gain practical experience in a professional setting relevant to their field of study. Through *magang*, students apply the knowledge and skills learned in the classroom to real-world projects, develop professional competencies, and build industry networks. The internship placement must be aligned with the Informatics study program—such as software development, artificial intelligence, cybersecurity, or digital interaction—and meet the duration and supervision requirements set by the university.

2. Kewirausahaan (Entrepreneurship & Innovation Program)

Students interested in entrepreneurship can join Skystar Ventures, UMN's tech-based business incubator. This program provides mentoring, workshops, and guidance from successful entrepreneurs and industry experts.

3. Riset (Research & Technology Program)

Students may engage in research projects, provided the research is approved by the Research and Innovation Support (RIS) Department of UMN and conducted under the supervision of an Informatics researcher involved in the project.

4. Proyek Independen (Road to Champion Program)

This program provides an opportunity for students to engage in independent projects, particularly those undertaken as part of academic competitions. Achievements in these competitions are formally recognized by the study program and may be converted into a minimum of 18 credits.

5. Proyek Desa (Community Engagement Program)

Students contribute to rural development through social and cultural initiatives, limited to activities approved by

the Community Engagement and Global Relations (CEGR) Department of UMN and conducted under the supervision of an Informatics lecturer involved in the project.

F. Distribution of Courses per Semester**Regular Path**

Semesters 1 to 4 are pre-assigned as a package and automatically enrolled for all students.

Semester 1							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 111	Ethics in Informatics	2	0		Compulsory	8	Yes
IF 120	Discrete Mathematics	3	0		Compulsory		
IF 130	Programming Fundamentals	3	0		Compulsory		Yes
CE 121	Linear Algebra	3	0		Compulsory		
CE 232	Digital Systems	3	0		Compulsory		
UM 122	English 1	2	0		Compulsory		
UM 142	Indonesian Language	2	0		Compulsory		
UM 152	Religion	2	0		Compulsory		
		20					

Semester 2							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 233	Algorithms & Data Structure	2	1	IF 120, IF 130, CE 121	Compulsory		
IF 260	Operating System	2	1		Compulsory		
CE 319	Probability and Statistics	3	0		Compulsory		
CE 332	Computer Architecture & Organization	3	0	CE 232	Compulsory		
EPM 101	Calculus	4	0		Compulsory		
UM 162	Pancasila	2	0		Compulsory		
UM 223	English 2	2	0	UM 122	Compulsory		
		20					

Semester 3							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 333	Formal Languages & Automata Theory	3	0	IF 120, IF 232	Compulsory		Yes
IF 350	Software Engineering & Project Management	3	0		Compulsory		
IF 351	Database Systems	2	1	IF 111	Compulsory		
IF 352	Basic Web Programming	2	1		Compulsory		
IF 440	Artificial Intelligence	3	0	EPM 101, CE 121, CE 319	Compulsory	3	Yes
UM 163	Civics	2	0		Compulsory		
MSC 3401	Public Speaking & Presentation Skills	3	0	UM 142, UM 223	Compulsory		
		20					

Semester 4							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 431	Algorithms & Complexity	3	0	IF 333	Compulsory		Yes
IF 433	Object Oriented Programming	2	1	IF 232	Compulsory		
IF 441	Machine Learning	3	0	IF 111, IF 440	Compulsory	3	Yes
IF 450	Human Computer Interaction	3	0	IF 111, IF 350	Compulsory		
IF 451	Advanced Web Programming	2	1	IF 351, IF 352	Compulsory		
IF 470	Computer Security	3	0	IF 260, IF 351, IF 352	Compulsory		
CE 449	Computer Network	2	0		Compulsory		
		20					

Semester 5							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 542	Deep Learning	3	0	IF 441	Compulsory	3	Yes
IF 570	Mobile Application Programming	2	1	IF 433, IF 450	Compulsory		
IF 571	Cybersecurity	3	0	IF 111, IF 470, CE 449	Compulsory		
IF 581	Game Development	2	1	IF 433, IF 450	Compulsory		
IF 590	Information Technology Research	2	0	IF 111, MSC 3401	Compulsory		
EM 105	Technopreneurship	3	0		Compulsory		
UM 321	English 3	2	0	UM 223	Compulsory		
		19					

Semester 6							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 6XY	MBKM	20	0	IF 111	Elective		
OR							
IF 670	Mobile Cross-Platform Programming	2	1	IF 433	Elective		
IF 675	Blockchain Technologies	2	0	IF 233, IF 260, IF 470	Elective	16	
IF 680	Multimedia Concept	3	0	IF 130	Elective		
IS 411	Data Modelling	1	1	IF 440	Elective		Yes
IS 412	Data Visualization	1	1	IF 440	Elective		
IS 513	E-Business	2	0	IF 111	Elective		
CE 542	System Administration	2	0	IF 260	Elective		
CE 739	Pervasive Computing	2	1	IF 260	Elective	Yes	
		19					

Semester 7							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 6XY	Internship Track 1	20	0	IF 111	Compulsory		
		20					

Semester 8							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 800	Final Project	6	0	IF 590	Compulsory		
		20					

G. Distribution of Courses per Semester

Acceleration Path

Semesters 1 to 4 follow the same structure as the Regular Path.

Semester 5							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 542	Deep Learning	3	0	IF 441	Compulsory	3	Yes
IF 570	Mobile Application Programming	2	1	IF 433, IF 450	Compulsory		
IF 571	Cybersecurity	3	0	IF 111, IF 470, CE 449	Compulsory		
IF 581	Game Development	2	1	IF 433, IF 450	Compulsory		
IF 590	Information Technology Research	2	0	IF 111, MSC 3401	Compulsory		
EM 105	Technopreneurship	3	0		Compulsory		
UM 321	English 3	2	0	UM 223	Compulsory		
Additional elective course(s)							
		>= 18					

Semester 6							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 6XY	Independent Project	18	0	IF 111	Elective		
IF 800	Final Project	6	0	IF 590	Compulsory		
		24					

Semester 7							
Course Code	Course Name	Credit		Prerequisite Course Code	Compulsory / Elective	SDG	AI
		Lect.	Lab.				
IF 6XY	Internship Track 1	20	0	IF 111	Compulsory		
		20					

H. Professional Certifications (CED UMN & Pearson)

1. IT Specialist: Artificial Intelligence

This certification focuses on the development of artificial intelligence (AI) solutions, including governance, transparency, security, and ethical considerations in AI. It is designed for individuals seeking to understand the AI development lifecycle, data processing, feature engineering, algorithm selection, model performance evaluation, and production pipeline management for continuous improvement.

2. IT Specialist: Cybersecurity

This certification covers the principles of cybersecurity, frameworks, and investigative skills. It is intended for aspiring security technicians or entry-level analysts who want to understand system vulnerabilities, analyze logs for incidents, and use monitoring tools for devices and networks.

3. IT Specialist: Software Development

This certification emphasizes core software development skills, including object-oriented programming, web applications, and database systems. It is suitable for individuals who wish to demonstrate their proficiency in C#, SQL, HTML, CSS, JavaScript, as well as programming concepts, JSON, XML, and foundational algebra.

4. Unity Certified User: Programmer

This certification validates programming competence using Unity, a leading platform for game and interactive application development. It focuses on foundational coding concepts, C# scripting, and feature implementation within the Unity environment. Ideal for those pursuing a career in game development or interactive media.

5. Unity Certified User: VR Developer

This certification emphasizes the ability to create VR programs using Unity in accordance with industry standards. Designed for individuals aiming to demonstrate expertise in VR development, it is well-suited for those pursuing careers in VR content creation and program management. A solid understanding of C#, VR software, and hardware is required.

In addition to the certifications mentioned above, students may take other professional certifications provided that:

1. The certification exam is proctored (supervised) to ensure its credibility.
2. The certification is in the field of Informatics—such as software development, artificial intelligence, cybersecurity, or digital interaction.
3. The certification is international and delivered in English.

Prior to registration, the Certification Approval Form available at CED UMN must be completed to ensure the certification is recorded and counted toward academic requirements.